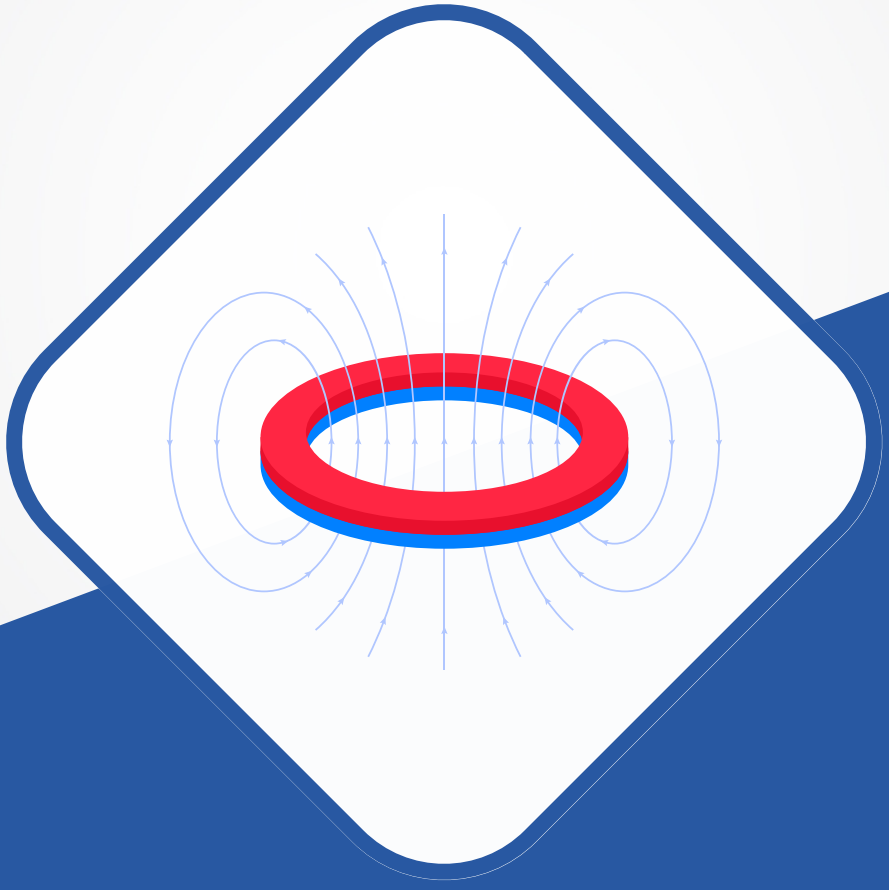
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Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Magnetic Effect of Current

English Medium

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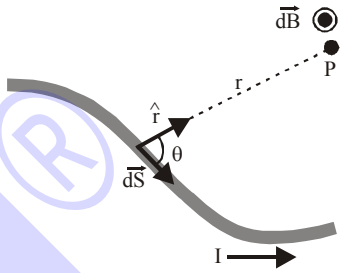
MAGNETIC EFFECT OF CURRENT

KEY CONCEPT

The branch of physics which deals with the magnetism due to electric current or moving charge (i.e. electric current is equivalent to the charges or electrons in motion) is called electromagnetism.

1. BIOT-SAVART LAW

The Biot-Savart law gives an expression for the magnetic field contribution, $d\vec{B}$, from the current source, $Id\vec{s}$,



$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{Id\vec{s}\times\hat{r}}{r^2}$$

where μ_0 is a constant called the permeability of free space:

$\mu_0 = 4\pi \times 10^{-7}$ T.m/A here Tesla (T) is SI unit of $\vec{B}$

Adding up these contributions to find the magnetic field at the point *P* requires integrating over the current source,

$$\vec{B} = \int d\vec{B} = \frac{\mu_0 I}{4\pi}\int\frac{d\vec{s}\times\hat{r}}{r^2}$$

- **Right hand thumb rule**

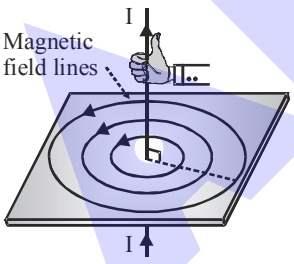
This rule gives the pattern of magnetic field lines due to current carrying wire.

(i) Straight current

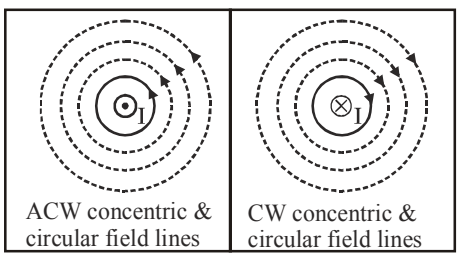
Thumb → In the direction of current

Curling fingers → Gives field line pattern

Case I : wire in the plane of the paper



Case II : Wire is ⊥ to the plane of the paper.

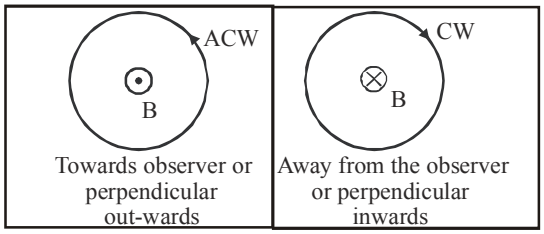


(ii) Circular current

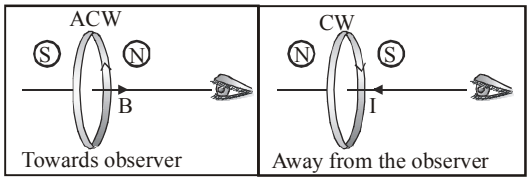
Curling fingers → In the direction of current,

Thumb → Gives field line pattern

Case I : wire in the plane of the paper

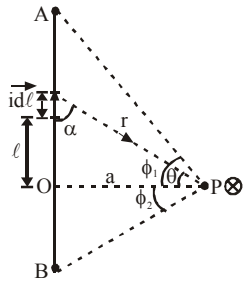


Case II : Wire is ⊥ to the plane of the paper



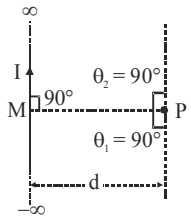
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- **Magnetic field due to a thin straight current carrying conductor**



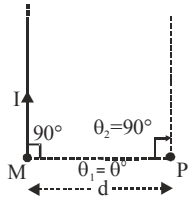
$$B = \frac{\mu_0 i}{4\pi a}\left[\sin\phi_1 + \sin\phi_2\right] \text{ (inwards)}$$

- **Magnetic field due to infinite length wire at point 'P'**



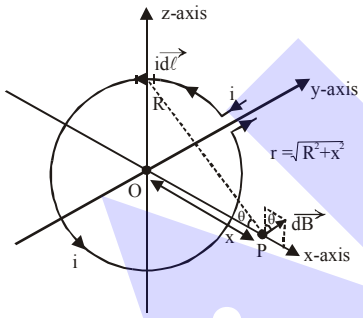
$$B_P = \frac{\mu_0 I}{2\pi d}$$

- **Magnetic field due to semi infinite length wire at point 'P'**



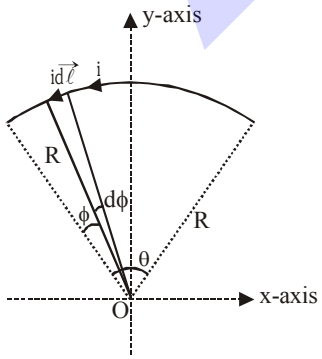
$$B_P = \frac{\mu_0 I}{4\pi d}$$

- **Magnetic field due to a current carrying loop**



$$B = \frac{\mu_0}{4\pi}\frac{i \times 2\pi R^2}{\left(R^2 + x^2\right)^{3/2}}$$

- **The magnetic field at the centre of a current carrying conductor bent in the form of an arc subtending angle θ at its centre.**



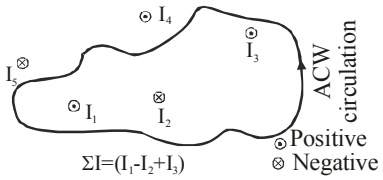
$$B = \frac{\mu_0 i}{4\pi R}\theta$$

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2. AMPERE'S CIRCUITAL LAW

Ampere's circuital law state that line integral of the magnetic field around any closed path in free space or vacuum is equal to μ_0 times of net current or total current which crossing through the area bounded by the closed path. Mathematically $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \Sigma I$



This law independent of size and shape of the closed path.

Any current outside the closed path is not included in writing the right hand side of law

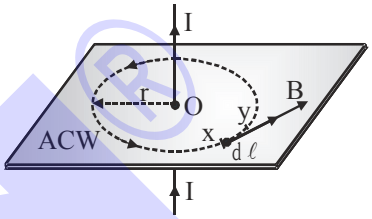
Note :

- This law suitable for infinite long and symmetrical current distribution.
- **Magnetic field due to infinite long thin current carrying straight conductor**

$\oint \vec{B} . d\vec{\ell} = \mu_0 \Sigma I$, $\oint B d\ell \cos\theta = \mu_0 I$ (where $\theta = 0°$)

$\oint B d\ell \cos 0° = \mu_0 I \Rightarrow B \oint d\ell = \mu_0 I$ (where $\oint d\ell = 2\pi r$)

$B\,(2\pi r) = \mu_0 I \Rightarrow B = \dfrac{\mu_0 I}{2\pi r}$



- **Magnetic field due to infinite long solid cylinderical conductor**

For a point inside the cylinder (r < R),

$B_{in}\,(2\pi r) = \mu_0 I' = \mu_0 \dfrac{I r^2}{R^2}$

$\Rightarrow B_{in} = \dfrac{\mu_0 I r}{2\pi R^2}$

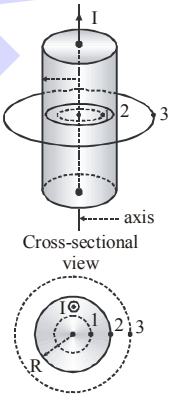
$\Rightarrow B_{in} \propto r$

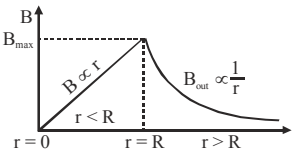
For a point outside the cylinder (r > R) :-

$B_{out}\,(2\,\pi\,r) = \mu_0 I$

$\Rightarrow B_{out} = \dfrac{\mu_0 I}{2\pi r}$

$\Rightarrow B_{out} \propto \dfrac{1}{r}$





- **Magnetic field due to infinite long hollow cylinderical conductor**

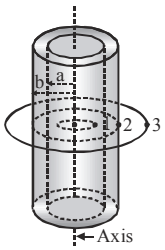
For a point at a distance r such that r < a < b $B_1 = 0$

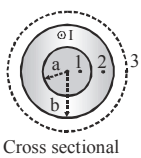
For a point at a distance r such that a < r < b

$B_2(2\pi r) = \mu_0 I' \Rightarrow B_2(2\pi r) = \mu_0 I\left(\dfrac{r^2 - a^2}{b^2 - a^2}\right)$

$B_2 = \dfrac{\mu_0 I}{2\pi r}\left(\dfrac{r^2 - a^2}{b^2 - a^2}\right)$ → r = a (inner surface) $\Rightarrow B_{IS} = 0$

→ r = b (outer surface) $\Rightarrow B_{OS} = \dfrac{\mu_0 I}{2\pi b}$ (maximum)



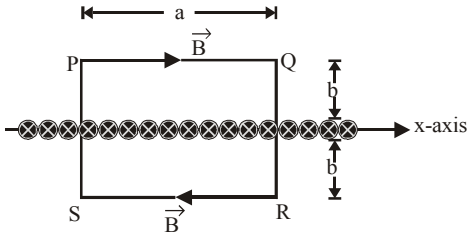


For a point at a distance r such that r > b > a, $B_3\,(2\pi r) = \mu_0 I \Rightarrow B_3 = \dfrac{\mu_0 I}{2\pi r}$

For a point at the axis of cylinder r = 0 $B_{axis} = 0$

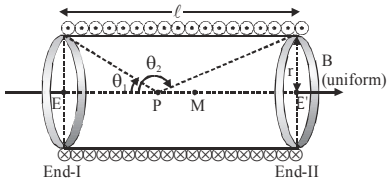
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Magnetic field due to an infinite plane sheet of current



$B = \frac{\mu_0 k}{2}$ (Where k is the current per unit length)

Magnetic field due to a finite solenoid



$B_P = \frac{\mu_0 nI}{2}(\cos\theta_1 - \cos\theta_2)$

Magnetic field due to a long solenoid

$B = \mu_0 ni$ (Where n is the number of turns per unit length)

3. FORCE ON A CHARGE PARTICLE IN ELECTRIC FIELD AND MAGNETIC FIELD

In the presence of both electric field $\vec{E}$ and magnetic field $\vec{B}$, the total force on a charged particle is

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

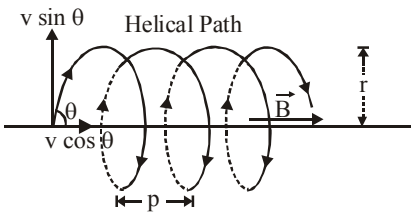
This is known as the Lorentz force.

4. MOTION OF A CHARGED PARTICLE IN A MAGNETIC FIELD

- **Case I : When the charged particle is moving perpendicular to the field.**

In this case the trajectory is circular as the force due to magnetic field on the charged particle is always perpendicular to velocity. As the work done by the magnetic field is zero the particle moves in the circular trajectory with constant speed. Radius $R = \frac{mv}{qB}$, $T = \frac{2\pi m}{qB}$

- **Case II : When the charged particle is moving at an angle θ to the field ($\theta \neq 0°, 90°$ or $180°$)**



The resultant path will be a helix with its axis parallel to the field $\vec{B}$ as shown in fig.

The pitch p of the helix = linear distance travelled in one rotation $p = T(v\cos\theta) = \frac{2\pi m}{qB}(v\cos\theta)$

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- **Velocity Selector :**

In the presence of magnetic field and electric field, particles which move with a certain velocity can be selected. This was the principle used by J. J. Thomson to measure the charge-to-mass ratio of the electrons. In Figure the schematic diagram of Thomson's apparatus is depicted.

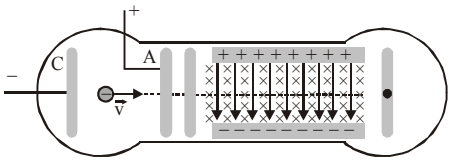


Figure : Thomson's apparatus

The electrons with charge $q = -e$ and mass m are emitted from the cathode C and then accelerated toward slit A. Let the potential difference between A and C be $V_A - V_C = \Delta V$. The change in potential energy is equal to the external work done in accelerating the electrons: $\Delta U = W_{ext} = q\Delta V = -e\Delta V$. By energy conservation, the kinetic energy gained is $\Delta K = -\Delta U = mv^2/2$. Thus, the speed of the electrons is given by

$$v = \sqrt{\frac{2e\Delta V}{m}}.$$

If the electrons further pass through a region where there exists a downward uniform electric field, the electrons, being negatively charged, will be deflected upward. However, if in addition to the electric field, a magnetic field directed into the page is also applied, then the electrons will experience an additional downward magnetic force $-e\vec{v}\times\vec{B}$. When the two forces exactly cancel, the electrons will move in a straight path. From Eq., we see that when the condition for the cancellation of the two forces is given by $eE = evB$. which implies

$$v = \frac{E}{B}$$

In other words, only those particles with speed $v = E / B$ will be able to move in a straight line. Combining the two equations, we obtain

$$\frac{e}{m} = \frac{E^2}{2(\Delta V)B^2}$$

By measuring E, ΔV and B, the charge-to-mass ratio can be readily determined. The most precise measurement to date is $e/m = 1.758820174(71) \times 10^{11}$ C/kg.

- **Mass Spectrometer :**

Various methods can be used to measure the mass of an atom. One possibility is through the use of a mass spectrometer. The basic feature of a *Bainbridge* mass spectrometer is illustrated in Figure. A particle carrying a charge $+q$ is first sent through a velocity selector. The applied electric and magnetic fields satisfy the relation $E = vB$ so that the trajectory of the particle is a straight line. Upon entering a region where a second magnetic field $\vec{B}_0$ pointing into the page has been applied, the particle will move in a circular path with radius r and eventually strike the photographic plate.

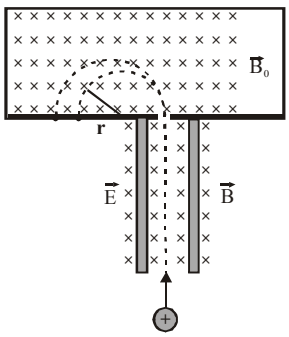


Figure : A Bainbridge mass spectrometer

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Using Eq., we have

$$r = \frac{mv}{qB_0}$$

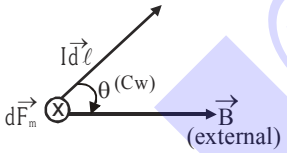
Since v = E/B, the mass of the particle can be written as

$$m = \frac{qB_0 r}{v} = \frac{qB_0 Br}{E}$$

5. MAGNETIC FORCE ON CURRENT CARRYING ELEMENT PLACED IN MAGNETIC FIELD

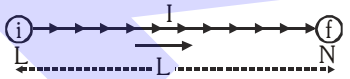
- **Magnetic force on current element**

$$d\vec{F}_m = I(d\vec{\ell} \times \vec{B})$$



- Total magnetic force on straight current carrying conductor in uniform magnetic field given as

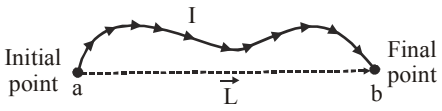
$$\vec{F}_m = \int_i^f d\vec{F}_m \left[\int_i^f d\vec{\ell} \right] = I \times \vec{B},\ \vec{F}_m = I(\vec{L} \times \vec{B})$$



Where $\vec{L} = \int_i^f d\vec{\ell}$, vector sum of all length elements from initial to final point, which is in accordance with the law of vector addition and $|\vec{L}|$ = length of the condutor.

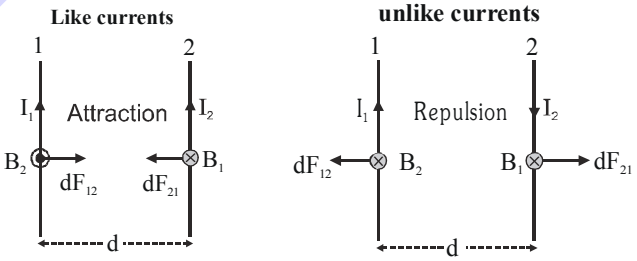
- Total magnetic force on arbitrary shape current carrying conductor in uniform magnetic field $\vec{B}$ is

$$\int_i^f d\vec{F}_m = I\left[\int_i^f d\vec{\ell} \right] \times \vec{B},\ \vec{F}_m = I(\vec{L} \times \vec{B})\ (L = ab)$$



Where $\vec{L} = \int_i^f d\vec{\ell}$, vector sum of all length elements from initial to final point or displacement between free ends of an arbitrary conducter from initial to final point.

Magnetic force between two parallel current carrying conductors

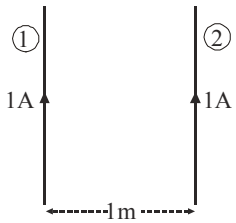


Magnetic force per unit length of each conductor is $\frac{dF_{12}}{d\ell} = \frac{dF_{21}}{d\ell} = \frac{\mu_0 I_1 I_2}{2\pi d}$

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Definition of ampere :

'Ampere' is the current which, when passed through each of two parallel infinite long straight conductors placed in free space at a distance of 1 m from each other, produces between them a force of 2×10^{-7} N/m



Torque on a current carrying loop placed in a magnetic field :

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

Where $\vec{\mu} = NI\vec{A}$ is called the magnetic moment vector of the loop.

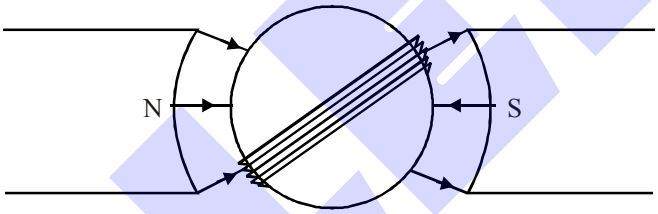
The above equation is analogous to $\vec{\tau} = \vec{p} \times \vec{E}$ in Eq., the torque exerted on an electric dipole moment $\vec{p}$ in the presence of an electric field $\vec{E}$.

Configuratoin energy of current loop in uniform magnetic field.

$U = -\mu B \cos\theta = -\vec{\mu} \cdot \vec{B}$

Moving coil Galvanometer :

The main parts of a moving-coil galvanometer are shown in figure.



The current to be measured is passed through the galvanometer. As the coil is in the magnetic field $\vec{B}$ of the permanent magnet, a torque $\vec{\Gamma} = ni\vec{A} \times \vec{B}$ acts on the coil. Here n = number of turns, i = current in the coil $\vec{A}$ = area-vector of the coil and $\vec{B}$ = magnetic field at the site of the coil. This torque deflects the coil from its equilibrium position.

The pole pieces are made cylindrical. As a result, the magnetic field at the arms of the coil remains parallel to the plane of the coil everywhere even as the coil rotates. The deflecting torque is then $\Gamma = niAB$. As the upper end of the suspension strip W is fixed, the strip gets twisted when the coil rotates. This produces a restoring torque acting on the coil. If the deflection of the coil is θ and the torsional constant of the suspension strip is k, the restoring torque is $k\theta$. The coil will stay at a deflection θ where

$$niAB = k\theta$$

or, $$i = \frac{k}{n\,AB}\theta$$

Hence, the current is proportional to the deflection. The constant $\frac{k}{n\,AB}$ is called the galvanometer constant.

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We define the **current sensitivity** of the galvanometer as the deflection per unit current. From Eq. this current sensitivity is.

$$\frac{\phi}{I}=\frac{NAB}{k}$$

A convenient way for the manufacturer to increase the sensitivity is to increase the number of turns N. We choose galvanometers having sensitivities of value, required by our experiment.

We define the **voltage sensitivity** as the deflection per unit volt of applied potential difference

$$\frac{\phi}{I}=\left(\frac{NAB}{k}\right)\frac{I}{V}=\left(\frac{NAB}{k}\right)\frac{1}{R}$$

An interesting point to note is that increasing the current sensitivity may not necessarily increase the voltage sensitivity. If $N \rightarrow 2N$, i.e., we double the number of turns, then

$$\frac{\phi}{I}\rightarrow 2\frac{\phi}{I}$$

Thus, the current sensitivity doubles. However, the resistance of the galvanometer is also likely to double, since it is proportional to the length of the wire. In eq. $N \rightarrow 2N$, and $R \rightarrow 2R$, thus the voltage sensitivity,

$$\frac{\phi}{V}\rightarrow\frac{\phi}{V}$$

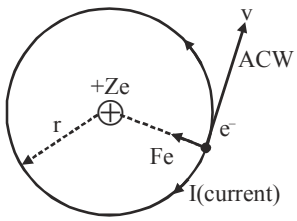
remains unchanged.

6. ATOMIC MAGNETISM

An atomic orbital electron, which doing bounded uniform circular motion around nucleus. A current constitues with this orbital motion and hence orbit behaves like current carrying loop. Due to this magnetism produces at nucleus position. This phenomenon called as 'atomic magnetism.

Bohr's postulates :

(i) $\frac{mv^2}{r}=\frac{kze^2}{r^2}$ (ii) $L = mvr = n\left(\frac{h}{2\pi}\right)$, where n = 1, 2,3 .....



Basic elements of atomic magnetism :

(a) Orbital current : $I = ef = \frac{e}{T} = \frac{ev}{2\pi r} = \frac{e\omega}{2\pi}$

(b) Magnetic induction at nucleus position : As circular orbit behaves like current carrying loop, so

magnetic induction at nucleus position $B_N=\frac{\mu_0 I}{2r}$

$$B_N=\frac{\mu_0 ef}{2r}=\frac{\mu_0 e}{2Tr}=\frac{\mu_0 ev}{4\pi r^2}=\frac{\mu_0 e\omega}{4\pi r}$$

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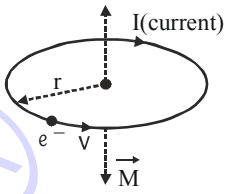
(c) Magnetic moment of circular orbit : Magnetic dipole moment of circular orbit

M = IA where A is area of circular orbit. $M = ef\left(\pi r^2\right) = \frac{\pi er^2}{T} = \frac{evr}{2} = \frac{e\omega r^2}{2}$

- **Relation between magnetic moment and angular momentum of orbital electron**

Magnetic moment $M = \frac{evr}{2} \times \frac{m}{m} = \frac{eL}{2m}$ ($\because$ angular momentum L = mvr)

Vector form $\vec{M} = \frac{-e\vec{L}}{2m}$



For orbital electron its $\vec{M}$ and $\vec{L}$ both are antiparallel axial vectors.

- **A nonconducting charged body is rotated with some angular speed.**

In this case the ratio of magnetic moment and angular momentum is constant which is equal to $\frac{q}{2m}$

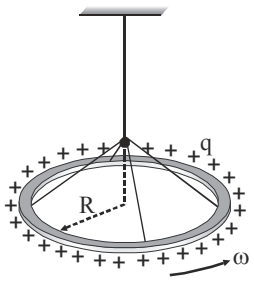
here q = charge and m = the mass of the body.

In case of a ring, of mass m, radius R and charge q distributed on it circumference.

Angular momentum $L = I\omega = (mR^2)(\omega)$ ... (i)

Magnetic moment $M = iA = (qf)(\pi R^2)$

$$M = (q)\left(\frac{\omega}{2\pi}\right)(\pi R^2) = q\frac{\omega R^2}{2} \text{ ...(ii)}$$



$\because f = \frac{\omega}{2\pi}$ From Eqs. (i) and (ii) $\frac{M}{L} = \frac{q}{2m}$

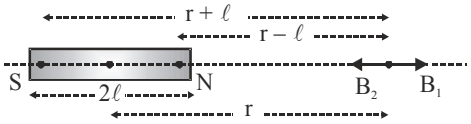
Although this expression is derived for simple case of a ring, it holds good for other bodies also. For example, for a disc or a sphere. $M = \frac{qL}{2m} \Rightarrow M = \frac{q(I\omega)}{2m}$, where $L = I\omega$

Rigid body	**Ring**	**Disc**	**Solid sphere**	**Spherical shell**
Moment of inertia (I)	mR^2	$\frac{mR^2}{2}$	$\frac{2}{5}mR^2$	$\frac{2}{3}mR^2$
Magnetic moment= $\frac{qI\omega}{2m}$	$\frac{q\omega R^2}{2}$	$\frac{q\omega R^2}{4}$	$\frac{q\omega R^2}{5}$	$\frac{q\omega R^2}{3}$

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SUPPLEMENT FOR JEE-MAINS

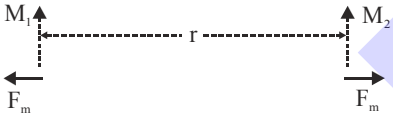
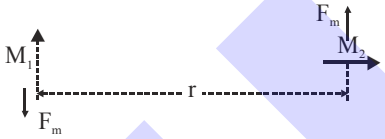
7. MAGNETIC FIELD DUE TO A BAR MAGNET



(i) At Axial position : $B_{axis} \simeq \frac{\mu_0}{4\pi}\frac{2M}{r^3}$

(ii) At equatorial position : $B_{eq.} \simeq \frac{\mu_0}{4\pi}\cdot\frac{M}{r^3}$

Dipole - Dipole Interactions :

S.No.	Relative position of dipoles	Magnetic force (F_m)
(a)		$\frac{\mu_0}{4\pi}\cdot\frac{6M_1M_2}{r^4}$ (along r)
(b)		$\frac{\mu_0}{4\pi}\cdot\frac{3M_1M_2}{r^4}$ (along r)
(c)		$\frac{\mu_0}{4\pi}\cdot\frac{3M_1M_2}{r^4}$ (perpendicular to r)

8. EARTH-MAGNETISM

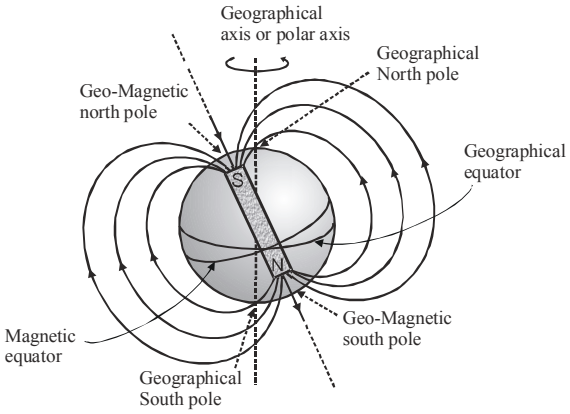
Important definitions :

(a) Geographic axis : It is a straight line passing through the geographical poles of the earth. It is also called axis of rotation or polar axis of the earth.

(b) Geographic Meridian (GM) : It is a vertical plane at any place which passing through geographical axis of the earth.

(c) Geographic equator : It is a great circle on the surface of the earth, in a plane perpendicular to the geographic axis. All the points on the geographic equator are at equal distance from the geographic poles.

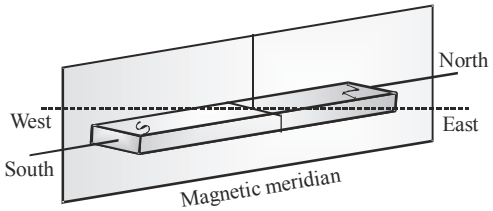
A great plane which passes through geographic equator and perpendicular to the geographic axis called **geographic equatorial plane**. This plane cuts the earth in two equal parts, a part has geographic north called **northen hemisphere** (NHS) and another part has geographic south called **southern hemi sphere** (SHS).



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(d) Magnetic axis : It is a straight line passing through magnetic poles of the earth. It is inclined to the geographic axis at nearly 11.3°.

(e) Magnetic Meridian (MM) : (i) It is a vertical plane at any place which passing through magnetic axis of the earth. (ii) It is a vertical plane at any place which passing through axis of free suspended bar magnet or magetic needle.
(iii) It is a vertical plane at any place which contains all the magnetic field lines of earth of that place.

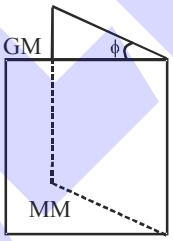


(f) Magnetic equator : It is a great circle on the surface of the earth, in a plane perpendicular to the magnetic axis. All the points on the magnetic equator are at equal distance from the magnetic poles.

Main elements of earth's magnetic field

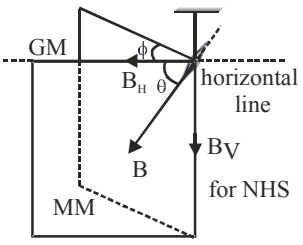
Angle of declination (ϕ)

At a given place the acute angle between geographic meridian and the magnetic meridian is called angle of declination, i.e. at a given place it is the angle between the geographical north south direction and the direction indicated by a agnetic compass needle in its equillibrium.



Angle of dip (θ)

(i) It is an angle which the direction of resultant magnetic field of the earth substends with the horizontal line in magnetic meridian at the given place.



(ii) It is an angle which the axis of freely suspended magnetic needle (up or down) substends with the horizontal line in magnetic meridian at a given place.

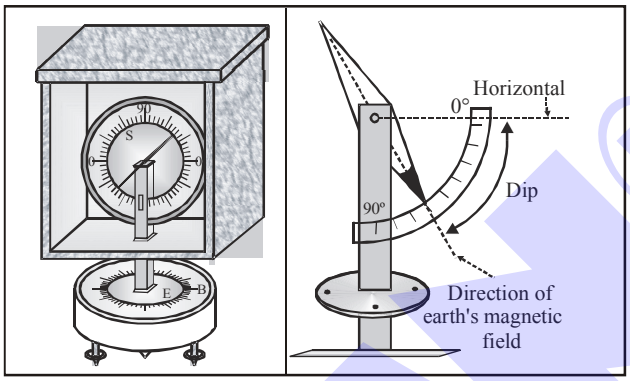
In northen hemi sphere, north pole of freely suspended magnetic needle will dip downwards i.e. towards the earth surface. In southern hemi sphere, south pole of freely suspended magnetic needle will dip downwards i.e. towards the earth surface.

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Dip circle : Angle of dip at a place is measured by the instrument called 'Dip-circle' in which a magnetic needle is free to rotate in vertical plane. About its horizontal axis. The ends of the needle move over a vertical scale graduated in degree.

Horizontal component of earth magnetic field (B_H)

Horizontal component of earth magnetic field at a given place is the component of resultant magnetic field of the earth along the horizontal line in magnetic meridian.



$B_H = B\cos\theta$ and $B_v = B\sin\theta$ .....(1)

so that $\tan\theta = \frac{B_V}{B_H}$ and $B = \sqrt{B_H^2 + B_V^2}$ .....(2)

At magnetic poles $\theta = 90°$ $B_H = 0$ and only B_V exist

❖ At magnetic equator $\theta = 0°$ $B_V = 0$ and only B_H exist

ϕ decides the plane in which magnetic field lies at any place, (ϕ) and (θ) decides the direction of magnetic field and (θ) and (B_H) decides the magnitude of the field.

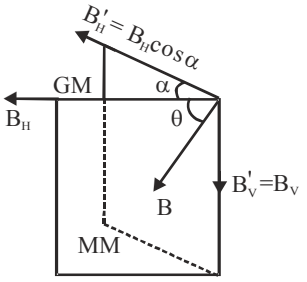
Apparent angle of dip (θ') :

When the plane of vertical scale of dip circle is in the magnetic meridian, the needle rest in the direction of earth's magnetic field. The angle made by the needle with the horizontal is called true dip or actual dip. If the plane of vertical scale of dip circle not kept in magnetic meridian, then the needle will not indicate the correct direction of earth magnetic field.

In this situation the angle made by the needle with the horizontal is called the apparent angle of dip. Suppose the dip circle is set at an angle α to the magnetic meridian. Effective horizontal component in this plane will be $B_H\cos\alpha$ and no effect on vertical component B_V

Apparent angle of dip $\tan\theta' = \frac{B'_V}{B'_H}$

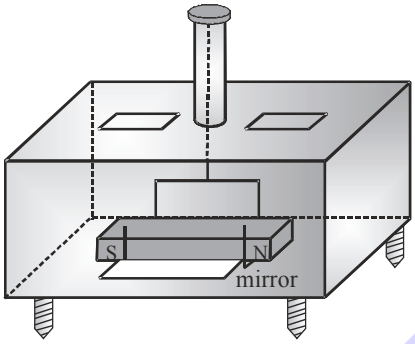
$\Rightarrow \tan\theta' = \frac{B_V}{B_H\cos\alpha} \Rightarrow \tan\theta' = \frac{\tan\theta}{\cos\alpha}$



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Vibration magnetometer :

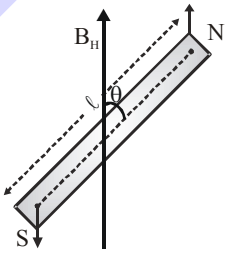
It is an instrument used to compare the horizontal components of magnetic field of earth of two different places, to compare magnetic fields and magetic moments of two bar magnets. It is also called oscillation magnetometer.



When a dipole is suspended in a uniform magnetic field it will align itself parallel to field. Now if it is given a small angular displacement θ about its equilibrium position. The restoring torque acts on it :

$\tau = -M B_H \sin\theta \Rightarrow I\alpha = -MB_H \sin\theta = -MB_H\theta, (\because \sin\theta \simeq \theta)$

$\Rightarrow \alpha = \frac{MB_H}{I}(-\theta) \Rightarrow \alpha = \omega^2(-\theta) \Rightarrow \omega^2 = \frac{MB_H}{I}$



The time period of angular S.H.M. $\Rightarrow T = 2\pi\sqrt{\frac{I}{MB_H}}$

M = magnetic moment of bar magnet

I = moment of inertia of bar magnet about its geometric axis

9. IMPORTANT DEFINITIONS AND RELATIONS

Magnetising field or Magnetic intensity $(\vec{H})$

Field in which a material is placed for magnetisation, called as magnetising field.

Magnetising field $\vec{H} = \frac{\vec{B}_0}{\mu_0} = \frac{\text{magnetic field}}{\text{permeability of free space}}$

SI Unit $\vec{H}$: ampere/meter

Intensity of magnetisation $(\vec{I})$

When a magnetic material is placed in magnetising field then induced dipole moment per unit volume of that material is known as intensity of magnetisation $\vec{I} = \frac{\vec{M}}{V}$

SI Unit : ampere/meter $\left[\because \frac{\vec{M}}{V} = \frac{I\vec{A}}{V} = \frac{\text{ampere} \times \text{meter}^2}{\text{meter}^3}\right]$

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Magnetic susceptibility (χ_m)

$$\chi_m = \frac{I}{H}$$ [It is a scalar with no units and dimensions]

Physically it represent the ease with which a magnetic material can be magnetised

A material with more χ_m, can be change into magnet easily.

Magnetic permeability μ

$$\mu = \frac{B_m}{H} = \frac{\text{total magnetic field inside the material}}{\text{magnetising field}}$$

It measures the degree to which a magnetic material can be penetrated (or permeated) by the magnetic field lines

Relative permeability $\mu_r = \frac{\mu}{\mu_0}$

It has no units and dimensions.

Relation between permeability and susceptibility

When a magnetic material is placed in magnetic field $\vec{B}_0$ for magnetisation then total magnetic field in material $\vec{B}_m = \vec{B}_0 + \vec{B}_i$, where $\vec{B}_i$ = induced field.

$\because \quad \vec{B}_0 = \mu_0 \vec{H}\,; \quad \vec{B}_i = \mu_0 \vec{I}$

$$\therefore \quad \vec{B}_m = \mu_0 \vec{H} + \mu_0 \vec{I} \Rightarrow \vec{B}_m = \mu_0(\vec{H} + \vec{I}) = \mu_0 \vec{H}\left(1 + \frac{I}{H}\right)$$

$$\Rightarrow \frac{B}{H} = \mu_0\left(1 + \frac{I}{H}\right) \Rightarrow \mu = \mu_0(1 + \chi_m) \Rightarrow \mu_r = 1 + \chi_m$$

for vacuum $\chi_m = 0, (\because \mu_r = 1)$

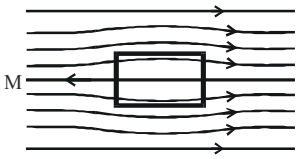
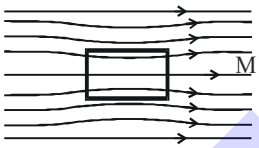
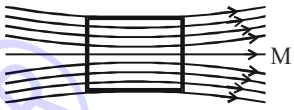
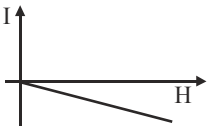
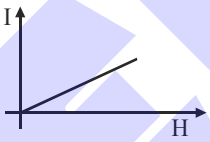
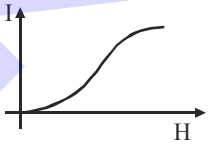
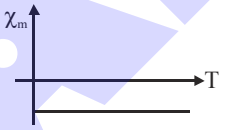
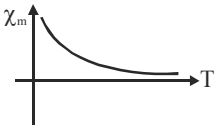
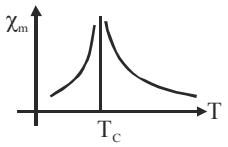
at STP for air $\chi_m = 0.04$

($\because$ at S.T.P. for air $\mu_r = 1.04$)

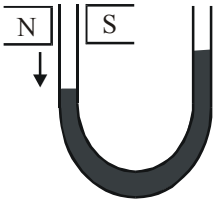
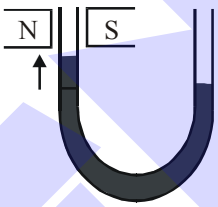
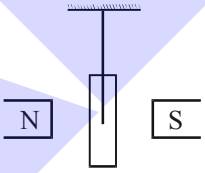
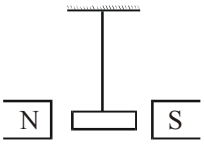
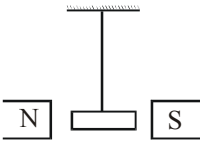
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10. CLASSIFICATION OF MAGNETIC MATERIALS

On the basis of magnetic properties of the materials [as magnetisation intensitily (I), Susceptibility (χ) and relative permeability (μ_r)] Faraday devide these materials in three classes –

PROPERTIES	DIAMAGNETIC	PARAMAGNETIC	FERROMAGNETIC
Cause of magnetism	Orbital motion of electrons	Spin motion of electrons	Formation of domains
Substance placed in uniform magnetic field.	Poor magnetisation in opposite direction. Here $B_m < B_0$ 	Poor magnetisation in same direction. Here $B_m > B_0$ 	Strong magnetisation in same direction. Here $B_m >>> B_0$ 
I – H curve	I → Small, negative, varies linearly with field 	I → Small, positive, varies linearly with field 	I → very large, positive & varies non–linearly with field 
χ_m – T curve	χ_m → small, negative & temperature independent $\chi_m \propto T^\circ$ 	χ_m → small, positive & varies inversely with temp. $\chi_m \propto \frac{1}{T}$ (Curie law) 	χ_m → very large, positive & temp. dependent $\chi_m \propto \frac{1}{T - T_C}$ (Curie Weiss law) (for $T > T_C$) (T_C = Curie temperature)  $T_C(I_{iron})$=770°C or 1043K
μ_r	$(\mu < \mu_0)$ $0 < \mu_r < 1$	$1 < \mu_r < 2$ $(\mu > \mu_0)$	$\mu_r >>> 1$ $(\mu >>> \mu_0)$
Magnetic moment of single atom	Atoms donot have any permanent magnetic moment	Atoms have permanent megnetic moment which are randomly oriented. (i.e. in absence of external magnetic field the magnetic moment of whole material is zero)	Atoms have permanent megnetic moment which are organised in domains.

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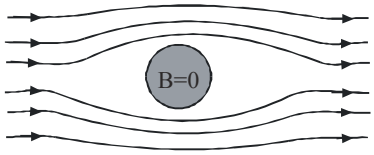
PROPERTIES	DIAMAGNETIC	PARAMAGNETIC	FERROMAGNETIC
Behaviour of substance in Nonuniform magnetic field	It moves from stronger to weaker magnetic field.  Weak Field  Strong Field  Level depressed in that limb	It moves with week force from weaker magnetic field to stronger magnetic field.  Weak Field  Strong Field  Level slightly rises	Strongly attract from weaker magnetic field to stronger magnetic field.  Weak Field  Strong Field
When rod of material is suspended between poles of magnet.	It becomes perpendicular to the direction of external magnetic field.  	If there is strong magnetic field in between the poles then rod becomes parallel to the magnetic field.  	Weak magnetic field between magnetic poles can made rod parallel to field direction. 
Magnetic moment of substance in presence of external magnetic field	Value $\vec{M}$ is very less and opposite to $\vec{H}$.	Value $\vec{M}$ is low but in direction of $\vec{H}$.	$\vec{M}$ is very high and in direction of $\vec{H}$.
Examples	Bi, Cu, Ag, Pb, H_2O, Hg, H_2, He, Ne, Au, Zn, Sb, NaCl, Diamond.(May be found in solid, liquid or gas).	Na, K, Mg, Mn, Sn, Pt, Al, O_2 (May be found in solid, liquid or gas.)	Fe,Co, Ni all their alloys, Fe_3O_4 Gd, Alnico, etc. (Normally found only in solids) (crystalline solids)

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Magnetic shielding

(i) Super conductors also provides perfect magnetic screening due to exclusion of field lines. This effect is called **'Meissner effect'**

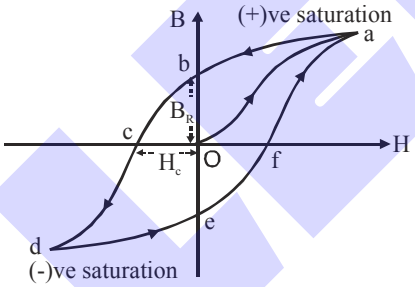
(ii) Relative magnetic premeability of super conductor is zero. So we can say that super conductors behaves like perfect dimagnetic.



Super conductor in a field

Magnetic Hysteresis

Only Ferromagnetic materials show magnetic hysteresis, when Ferromagnetic material is placed in external magnetic field for magnetisation then B increases with H non-linearly along Oa. If H is again bring to zero then it decreases along path ab. Due to lagging behind of B with H this curve is known as hysteresis curve. [Lagging of B behind H is called hysteresis]



Cause of hysteresis : By removing external magnetising field (H = 0), the magnetic moment of some domains remains aligned in the applied direction of previous magnetising field which results into a residual magnetism.

- **Residual magnetism (ob) = B_r** ≡ retentivity ≡ remanence
  Retentivity of a specimen is a measure of the magnetic field remaining in the ferromagnetic specimen when the magnetising field is removed.
- **Coercivity (oc) :** Coercivity is an measure of magnetising field required to destroy the residual magnetism of the ferromagnetic specimen.

Ferromagnetic materials

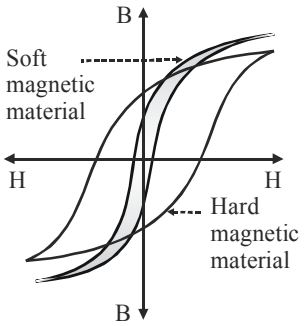
Soft magnetic materials	**Hard magnetic materials**
Low retentivity, low coercivity and small hysteresis loss. suitable for making electromagnets, cores of transformers etc. Ex. Soft iron, (used in magnetic shielding)	High retentivity, high coercivity and large hysteresis loss suitable for permanent magnet Ex. Steel, Alnico

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Hysteresis loss

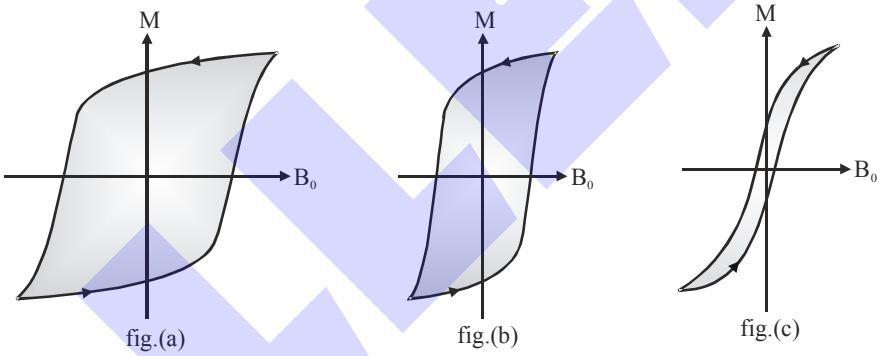
(i) The area of hystersis loop for a ferromagnetic material is equal to the energy loss per cycle of magnetisation and demagnetisation per unit volume.

$$W_H = \oint B.dH = \mu_0 \oint I.dH$$



(ii) Its value is different for different materials.

(iii) The work done per cycle per unit volume of material is equal to the area of hysteresis loop.



The materials of both (a) and (b) remain strongly magnetized when B_0 is reduced to zero. The material of (a) is also hard to demagnetize, it would be good for permanent magnets.

The material of (b) magnetizes and demagnetizes more easily, it could be used as a computer memory material.

The material of (c) would be useful for transformers and other alternating-current devices where zero hysteresis would be optimal.

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